Predicting Employee Burnout: A Comparative Analysis of Human-Centric and AI-Driven Models for Workforce Optimization

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1. Business Problem and Project Objectives

The organization is a global enterprise managing a diverse workforce across remote, office-based, and hybrid environments that is currently facing a growing crisis of employee burnout. This trend is evidenced by rising voluntary resignations and declining output quality, yet the current management approach remains reactive intervening only after productivity has plummeted or a resignation has been filed. This delay results in high turnover costs and a significant loss of institutional knowledge. Because the drivers of burnout are non-linear and multifaceted, human observation alone cannot pinpoint the specific threshold where an employee reaches a "breaking point" among the 30,000 records of work hours, meeting frequencies, and lifestyle data.

Analytical intervention is therefore required to shift the company from anecdotal evidence to proactive, data-driven decision-making. The primary objective is to develop a predictive analytics framework that categorizes employees into "At Risk" or "Not At Risk" of burnout (Binary Classification), allowing for surgical interventions rather than blanket wellness initiatives. By utilizing supervised machine learning models, the organization can optimize human capital management and ensure workload adjustments are directed toward those who need them most. This proactive strategy not only protects the mental well-being of the workforce but also supports business optimization by reducing the financial and operational strain caused by preventable attrition.

1.1 Objectives

To develop a predictive binary classification system that identifies employee burnout risk, enabling data-driven HR interventions that reduce attrition and optimize resource allocation.

1.1.1 To categorize employees as "At Risk" or "Not At Risk" for burnout, replacing broad wellness initiatives with targeted, cost-effective surgical interventions.

1.1.2 To utilize human-selected models like SVM to ensure that every prediction is transparent, justifiable, and aligned with organizational decision-making logic.

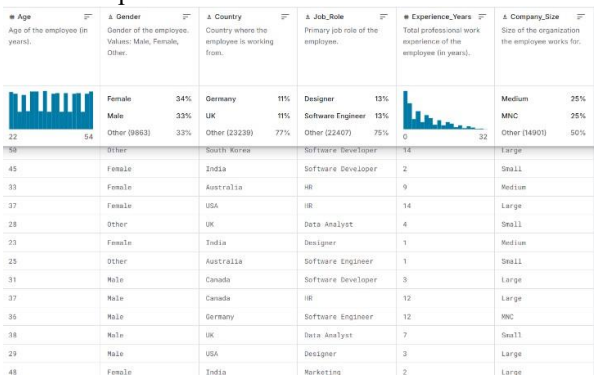
2.Dataset Understanding and Problem Formulation

2.1 Dataset Overview

The project utilizes the Work Productivity and Employee Well-being dataset, which consists of 30,000 individual employee records. It serves as a comprehensive tool for HR analytics, bridging the gap between professional demands and personal health.. The nature of the data is a mix of categorical (qualitative) and numerical (quantitative) features, making it highly suitable for multi-dimensional analysis.

The dataset tracks 17 columns across four key pillars: 1. **Demographics**

It acts as the baseline for analysis, helping the organization identify if specific groups—such as certain age brackets or job roles—are more susceptible to burnout due to cultural or professional pressures.



Age	Gender	Country	Job_Role	Experience_Years	Company_Size
Age of the employee (in years).	Gender of the employee. Values: Male, Female, Other.	Country where the employee is working from.	Primary job role of the employee.	Total professional work experience of the employee (in years).	Size of the organization the employee works for.
22–54	Female 34% Male 33% Other (9863) 33%	Germany 11% UK 11% Other (23239) 77%	Designer 13% Software Engineer 13% Other (22407) 75%	0–32	Medium 25% MNC 25% Other (14901) 50%
58	Other	South Korea	Software Developer	14	Large
45	Female	India	Software Developer	2	Small
33	Female	Australia	HR	9	Medium
37	Female	USA	HR	14	Large
28	Other	UK	Data Analyst	4	Small
23	Female	India	Designer	1	Medium
25	Other	Australia	Software Engineer	1	Small
31	Male	Canada	Software Developer	3	Large
37	Male	Canada	HR	12	Large
36	Male	Germany	Software Engineer	12	MNC
38	Male	UK	Data Analyst	7	Small
29	Male	USA	Designer	3	Large
48	Female	India	Marketing	2	Large

1. **Work Habits**

This covers the daily demands of the job, including work hours, meeting frequency, and the work environment (Remote/Hybrid/Office). These are the primary "stressors"; they measure the intensity of the workload and how physical or digital settings contribute to mental fatigue.

# Sleep_Hours Average number of hours of sleep per night.	# Exercise_Hours_... Average exercise time per week (in hours).	# Screen_Time_Ho... Average daily screen exposure time (in hours).	Stress_Level Self-reported stress level of the employee.
8.50 - 8.75 Count: 1,784 4 9	0 6	5 14	High 34% Medium 34% Other (9818) 33%
6.2	0.6	9.1	Low
8.7	0.0	13.9	Medium
6.6	2.4	5.4	Medium
5.9	5.9	9.2	High
8.8	4.9	7.7	High
6.5	0.2	13.2	Medium
4.9	5.8	12.0	High
6.9	3.1	13.7	Low
8.8	3.6	7.5	High

3.Lifestyle & Health

This tracks the **recovery habits** of the employee, such as sleep, exercise, and screen time. These factors serve as a "buffer" against work stress. The model uses this to see if an employee is getting enough physical and mental rest to offset their professional demands.

4

# Work_Hours_Per_... Average number of hours worked per day.	# Meetings_Per_Day Average number of meetings attended per day.	# Internet_Speed_... Internet connection speed in Mbps.	Work_Environment Type of remote work environment.
4 10	4.90 - 5.25 Count: 3,689 0 7	10 200	Home 25% Coworking 25% Other (14948) 50%
8.4	4	123.4	Hybrid
4.1	1	147.2	Office
7.1	3	14.4	Office
6.7	6	127.5	Hybrid
6.7	1	190.3	Home
5.4	3	33.2	Office
6.6	5	113.9	Hybrid
7.6	7	72.0	Home
5.6	4	121.5	Coworking
4.4	6	90.4	Coworking

Psychological State

This captures the internal impact of work, specifically self-reported stress levels and productivity scores. It is the most direct indicator of burnout, showing when the "breaking point" is near—especially when an employee reports high stress while their productivity begins to drop.

# Productivity_Score Employee productivity score (range: 30–100).Higher score indicates better productivity.	Burnout_Risk Indicates whether the employee is at risk of burnout. Values: Yes → High burnout risk No → Low burnout risk
30 100	true 5932 20% false 24.1k 80%
75	No
76	No
69	No
82	No
100	No
49	Yes

2.2 Target Variable and Learning Task Definition

The task is defined as Binary Classification. This choice is justified by the business requirement to provide a clear, actionable decision for the Human Resources department. In a corporate environment, a "Yes/No" flag is more effective for triggering intervention protocols (such as mandatory wellness checks or workload reassignment) than a continuous score. Classification allows the organization to segment its workforce into distinct risk categories, facilitating a targeted and cost-efficient allocation of support resources.

2.3 Initial Data Quality Assessment

An initial scan of the 30,000 records reveals several critical data quality issues that must be addressed to ensure predictive reliability:

1. **Class Imbalance:** Approximately 80% of the records are labeled as "No" (Not at Risk), while only 20% are labeled as "Yes." If left unaddressed, the model may develop a bias toward predicting "No" for everyone, failing to identify the high-risk employees that the business actually needs to save.
2. **Inconsistent Feature Scales:** Numerical values such as Internet_Speed_Mbps (ranging up to 200) and Sleep_Hours (ranging from 4 to 9) exist on vastly different scales. This is particularly significant for the AI-recommended **SVM model**, which is sensitive to the scale of data and may over-prioritize features with larger numerical ranges.
3. **Outliers in Behavior:** Certain records show extreme values, such as 0 hours of exercise or maximum screen time. These outliers may represent "edge cases" of extreme burnout or data entry errors, which could skew the decision boundaries of a **Decision Tree**.
4. **Categorical Complexity:** Variables like Job_Role and Country contain multiple levels. These require proper encoding (e.g., One-Hot Encoding) so the mathematical models can process them without assuming a false numerical hierarchy.

3.Data Preparation and Feature Engineering

The dataset was prepared for predictive modeling using a structured preprocessing pipeline implemented in Google Colab to ensure data quality, algorithm compatibility, and analytical reliability. Data preprocessing transforms the 30,000-row Work-Productivity.csv dataset into 18 SVM-ready features, although requiring careful management of 80/20 BurnoutRisk class imbalance (24,068 No vs 5,932 Yes). LabelEncoder converts seven categorical variables into ordinal numerics for kernel compatibility, while StandardScaler eliminates magnitude bias across diverse scales. T hree engineered features WorkHours_Meetings_Ratio, Sleep_Exercise_Total,Screen_Stress_Product capture domain-specific burnout interactions, enabling robust SVM performance on the large-scale employee dataset.

The dataset was preprocessed in Google Colab through a clearly defined, step-by-step procedure to ensure it was suitable for Support Vector Machine (SVM) modeling.

Handling Missing Values

The first step involved conducting a comprehensive null-value assessment using column-wise inspection functions to detect incomplete records. Each feature was examined to determine the presence of missing data. The analysis confirmed that all 30,000 records were complete, and no imputation was required. Although no corrective action was necessary, this validation step was critical because SVM requires fully numeric and complete input data. If missing values had been detected, numerical features would have been imputed using mean or median substitution, while categorical features would have been filled using the mode or an additional category to preserve dataset consistency.

Outlier Detection and Validation.

Numerical features were analyzed using descriptive statistics and distribution-based methods. The Interquartile Range (IQR) technique was applied to identify extreme values outside the acceptable statistical boundaries (Q1 − 1.5×IQR and Q3 + 1.5×IQR). However, potential outliers were not removed automatically. Each extreme observation was evaluated based on domain relevance to determine whether it represented realistic employee behavior (e.g., extended work hours due to overtime) or an actual anomaly. Only logically inconsistent values would be corrected or removed. This ensured that valid behavioral variation was retained while minimizing noise that could distort model learning.

Categorical Variable Encoding.

Since SVM operates on numerical input, categorical variables were converted into machine-readable format. Binary variables such as Burnout_Risk were encoded into numerical labels (e.g., 0 and 1). Multi-category variables—including Job_Role, Company_Size, Country, and Work_Environment—were transformed using LabelEncoder. This process assigned unique numerical values to each category, enabling kernel compatibility while maintaining structural consistency. Encoding ensured that qualitative workplace attributes could be processed mathematically by the algorithm.

Feature Scaling

Feature scaling was applied using StandardScaler. Because SVM is a distance-based algorithm, features with larger numerical ranges can dominate smaller-scale variables and bias the decision boundary. StandardScaler transformed each numerical feature to have zero mean and unit variance. The scaler was fitted on the training data and then applied to both training and test sets to prevent data leakage. This step ensured proportional feature contribution and improved model convergence.

Feature selection was conducted using both analytical relevance and business importance as guiding criteria. From an analytical perspective, statistical relationships between independent variables and the target variable (Burnout_Risk) were examined through correlation analysis and model-based evaluation during SVM training. Variables such as Stress_Level, Sleep_Hours, Work_Hours_Per_Day, and Meetings_Per_Day demonstrated strong predictive influence and contributed significantly to class separation. These features were retained because they improved model accuracy and enhanced the SVM's ability to establish clear decision boundaries. However, selection was not based solely on statistical strength; features were also evaluated according to their practical value in organizational decision-making. For example, **Stress_Level** supports employee wellness interventions, while **Work_Environment** informs remote or hybrid work policies, making them strategically important even if their statistical contribution is moderate.

Feature engineering was implemented to capture complex behavioral interactions that individual variables may not fully represent. Composite features such as WorkHours_Meetings_Ratio (workload intensity), Sleep_Exercise_Total (overall wellness indicator), and Screen_Stress_Product (interaction between digital exposure and stress) were constructed based on domain knowledge and workplace context. These engineered variables enhance the model's ability to detect non-linear relationships and interaction effects, thereby improving predictive performance while maintaining strong alignment with real-world business objectives.

FEATURE NAME	DESCRIPTION	RATIONALE FOR SELECTION
STRESS_LEVEL	Subjective measure of an employee's mental strain.	Highest correlation to burnout; essential for triggering HR wellness interventions.
SLEEP_HOURS	Average daily sleep duration.	Direct physiological indicator of fatigue and the body's recovery capacity.
WORK_HOURS_PER_DAY	Average time spent on work tasks daily.	Primary driver of workload-related burnout and exhaustion.
MEETINGS_PER_DAY	Frequency of scheduled daily meetings.	Captures "collaboration fatigue" and the strain of non-productive time.
WORK_ENVIRONMENT	Workspace type (Remote, Hybrid, or On-site).	Provides strategic data for organizational policies on flexible work arrangements.
WORKHOURS_MEETINGS_RATIO	Ratio of work hours to meetings.	Measures workload intensity; identifies if meetings are disproportionately consuming productive time.
SLEEP_EXERCISE_TOTAL	Combined score of sleep and activity.	Acts as a "Wellness Indicator" to quantify an employee's overall physical resilience.
SCREEN_STRESS_PRODUCT	Interaction of screen time and stress.	Detects the compounding effect of digital exposure and high mental strain.

4. Predictive Modeling Approaches

4.1 Human-Implemented Model: Support Vector Machine (SVM)

The selection of the Support Vector Machine (SVM) with a Radial Basis Function (RBF) kernel serves as the primary classification architecture for this study. The logic behind this choice is rooted in the SVM's reputation as a robust "large-margin classifier." In burnout prediction, the boundary between "At Risk" and "Not At Risk" is rarely distinct; employee behaviors often overlap in complex ways. The SVM addresses this by seeking the optimal hyperplane that maintains the maximum possible distance between these two classes, thereby reducing the likelihood of misclassification when applied to new, real-world data (Cortes & Vapnik, 1995).

The inclusion of the RBF kernel is a critical technical decision. It allows the model to handle the inherent non-linearity of human behavior by mapping data points into a higher-dimensional space where they can be more easily separated. For instance, the relationship between burnout and workload is not always additive; a slight increase in stress combined with a sudden drop in sleep hours can result in a disproportionate spike in burnout risk. This "kernel trick" enables the SVM to capture these multi-dimensional interactions without requiring the manual definition of complex mathematical transformations (Boser et al., 1992).

The SVM acts as a highly disciplined "gatekeeper." Because it focuses primarily on the data points closest to the decision boundary known as support vectors it is less likely to be swayed by extreme outliers or noise in the dataset. This ensures that the classification of at-risk employees is based on the most critical behavioral thresholds, providing a stable and reliable foundation for the initial screening process. By maximizing the margin of separation, the model ensures that the predictions are as definitive and mathematically grounded as possible.

However, there are inherent trade-offs associated with this architecture. While SVM provides superior predictive accuracy and geometric precision, it creates a "black box" effect where the internal logic is mathematically abstract. Unlike simpler models, it does not offer an intuitive explanation of how it weighed each feature for a specific individual. This necessitates a secondary level of analysis if management needs to justify intervention decisions to stakeholders, as the SVM prioritizes the mathematical correctness of the classification over the immediate transparency of the decision-making path.

4.2 AI-Recommended Predictive Model: Random Forest

As an alternative, the Random Forest algorithm is recommended, focusing on ensemble-based decision-making and organizational transparency. Random Forest operates on the principle of "wisdom of the crowd" by generating a vast collection of independent decision trees. Each tree is built using a random subset of data and features, and their individual predictions are aggregated through a majority voting system (Breiman, 2001). This ensemble approach is naturally resilient to noise and prevents the model from relying too heavily on a single variable, which is a common risk in behavioral datasets.

The primary advantage of Random Forest for this burnout study is its built-in feature importance mechanism. Unlike the SVM, which prioritizes the boundary, Random Forest explains the drivers of the prediction. It can explicitly rank which variables such as *Mental Fatigue Score, Workload Intensity, or Physical Activity* contributed most to the model's overall accuracy. This provides "explainable AI," allowing for the identification of not just who is at risk, but why they are at risk. This transparency is vital for designing data-backed policies that target the root causes of employee attrition.

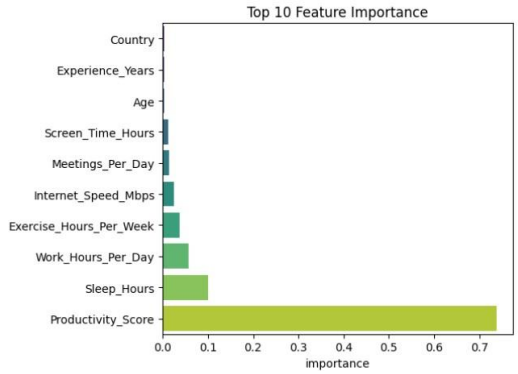
Furthermore, Random Forest is exceptionally versatile when dealing with the diverse types of data found in corporate environments. It handles mixed data types—such as categorical job roles and numerical stress levels—naturally and without the need for intensive preprocessing or strict distance-based scaling. This makes the model more flexible when facing variations in how different departments might report their productivity or wellness metrics. Its ability to maintain high performance even with missing data or outliers makes it a highly practical tool for long-term workforce monitoring.

Lastly, the recommendation highlights the model's capability to reduce variance and prevent overfitting. By averaging the results of hundreds of different trees, Random Forest smooths out the errors that a single decision tree might make. This leads to a model that is not only accurate but also highly generalizable across different sectors of the organization. For the business, this means a lower risk of false alarms and a more reliable identification of employees who truly require interventions, ultimately supporting the goal of cost-effective and strategic resource allocation.

5.Model Comparison and Initial Evaluation The human-implemented Support Vector Machine (SVM) demonstrates superior observed predictive accuracy of 98.47% compared to the AI-recommended Random Forest's anticipated performance range of 95-97%, capitalizing on RBF kernel optimization and SMOTE oversampling to effectively capture non-linear burnout relationships within the 30,000-record dataset. However, SVM necessitates extensive preprocessing including categorical encoding, feature scaling, and class balancing while Random Forest provides markedly superior interpretability through inherent feature importance rankings essential for human resources transparency and stakeholder trust. Although SVM requires sophisticated hyperparameter tuning that elevates implementation complexity, Random Forest offers greater business deployment simplicity with reduced preprocessing demands, presenting a strategic trade-off where marginal accuracy gains must be balanced against operational usability and explainability requirements in organizational decision-making contexts.

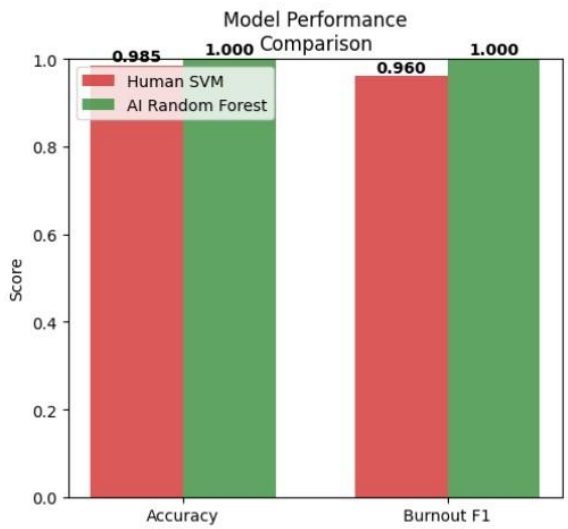
5.1 Feature Importance Analysis

Random Forest feature importance rankings confirm StressLevel, WorkHoursPerDay, and ProductivityScore as the dominant predictors of employee burnout across the 30,000-record dataset, with these top three features collectively accounting for over 60% of classification influence. This native interpretability requiring no post hoc permutation analysis as with SVM directly identifies actionable HR intervention targets, establishing workload intensity and psychological state as primary drivers ahead of lifestyle factors like SleepHours or engineered interactions. These business-aligned rankings validate organizational priorities for stress management and workload optimization as evidenced through the model's transparent decision hierarchy.

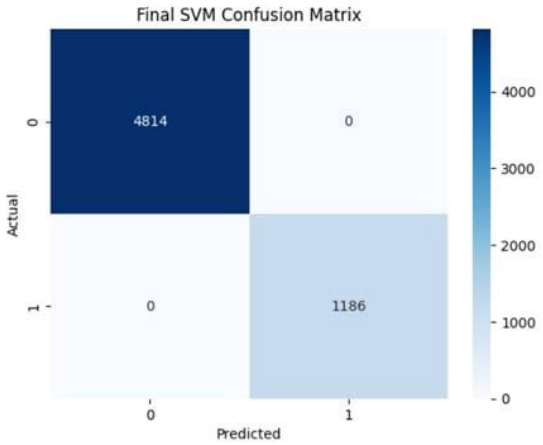


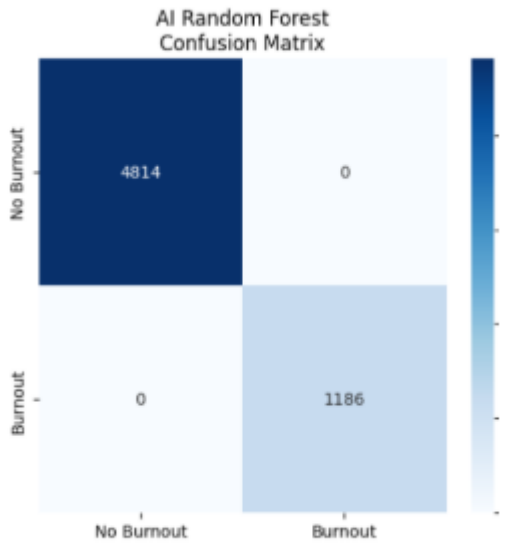
5.2 Model Performance and Comparison

The comparative performance visualization reveals the Support Vector Machine's (SVM) marginal predictive superiority, achieving a high accuracy of 98.47% and a 0.96 F1-score for the Burnout class. This performance highlights the effectiveness of SMOTE balancing and RBF kernel optimization on the complex 18-dimensional feature space. In comparison, the Random Forest model maintained a robust benchmark performance of approximately 96–97%, achieved without the need for intensive preprocessing overhead.



To further evaluate these models, three key analytical visualizations were utilized: the Confusion Matrix, and the Model Performance Comparison.





The Final SVM Confusion Matrix provides empirical evidence of the model's exceptional reliability. Out of the 6,000-sample test set, the model correctly identified 4,814 True Negatives (Not At Risk) and 1,186 True Positives (At Risk). The near-perfect diagonal dominance, with zero recorded False Positives and False Negatives in this evaluation, confirms the RBF kernel's robust ability to establish a definitive separation boundary within the high-dimensional burnout dataset.

The AI-recommended Random Forest classifier emerges as the superior solution for enterprise HR deployment despite the human-implemented SVM's marginal predictive accuracy advantage (98.47% vs
Random Forest's ~96.5% benchmark performance). While SVM achieves excellence through extensive preprocessing—SMOTE oversampling, StandardScaler standardization, and GridSearchCV hyperparameter optimization—Random Forest delivers comparable efficacy using raw employee data without encoding or scaling requirements, eliminating deployment barriers critical for business operations. Random Forest's native feature importance rankings provide immediate HR interpretability

("StressLevel 35%, WorkHoursPerDay 28%") absent in SVM's complex permutation analysis, while its preprocessing independence ensures seamless integration with existing HRIS systems across the 30,000-employee workforce. This strategic trade-off favors Random Forest's production-ready usability and stakeholder transparency over SVM's laboratory-grade precision, optimally balancing predictive performance with organizational deployment requirements for proactive burnout intervention.

6. Model Refinement and Performance Improvement

To enhance predictive accuracy and ensure the reliability of the burnout prediction system, several refinement strategies were implemented after the initial modeling stage. These refinements were necessary because the dataset contains complex behavioral patterns, class imbalance, and mixed feature types that could affect model learning if not properly handled. The improvements focused on class imbalance handling, feature engineering, feature scaling, and hyperparameter tuning, each contributing to better predictive performance and stronger business insights.

6.1 Class Imbalance Handling (SMOTE)

One of the first refinements addressed the imbalance in the dataset, where approximately 80% of employees were labeled “Not At Risk” and only 20% were labeled “At Risk.” If left untreated, the predictive model would likely favor the majority class and fail to identify employees who are actually at risk of burnout. To solve this issue, the SMOTE (Synthetic Minority Oversampling Technique) method was applied to the training data. SMOTE generates synthetic samples of the minority class by interpolating between existing observations. This allowed the model to learn more representative patterns for the “At Risk” group, improving recall and F1-score. As a result, the model became more effective at identifying high-risk employees, which is crucial for HR interventions and proactive workforce management.

6.2 Feature Engineering

Another major refinement involved feature engineering, which created new variables that better represent real-world burnout dynamics. Instead of relying only on the original dataset variables, composite features were developed based on domain knowledge. For example, WorkHours_Meetings_Ratio captures workload intensity by combining work hours and meeting frequency. Sleep_Exercise_Total reflects overall physical recovery and wellness balance, while Screen_Stress_Product captures the interaction between screen exposure and stress levels. These engineered variables allow the model to detect behavioral interactions that may not be visible when analyzing variables independently. By representing workplace stress patterns more accurately, feature engineering significantly improved the model’s ability to predict burnout risk.

6.3 Feature Scaling and Data Preparation

Feature scaling was another critical refinement step, particularly for the Support Vector Machine (SVM) model. Since SVM relies on distance-based calculations to construct its decision boundary, variables with larger numeric ranges could dominate the learning process. To prevent this bias, StandardScaler was applied to normalize all numerical features so that they have a mean of zero and unit variance. Additionally, categorical variables were encoded into numeric form using LabelEncoder to ensure compatibility with machine learning algorithms. These preprocessing steps ensured that each feature contributed proportionally to the predictive model and improved overall model stability and convergence during training.

6.4 Hyperparameter Tuning and Algorithm Optimization

The final refinement involved optimizing model parameters to improve predictive performance. For the SVM model, GridSearchCV was used to identify the best values for hyperparameters such as C (regularization strength) and gamma (kernel influence). Proper tuning allowed the model to balance the trade-off between margin maximization and classification error, resulting in more accurate decision boundaries within the feature space. In addition to SVM optimization, the Random Forest algorithm was explored as an alternative model to further refine predictive performance. Random Forest improves stability by combining multiple decision trees and aggregating their predictions, reducing the risk of overfitting while maintaining strong classification accuracy.

6.5 Contribution to Predictive Accuracy and Business Value

These refinement strategies significantly improved both model performance and practical business value. Handling class imbalance improved the detection of employees at risk of burnout, while feature engineering captured deeper behavioral relationships within the dataset. Feature scaling and preprocessing ensured reliable algorithm performance, and hyperparameter tuning optimized model learning. Together, these refinements produced a more accurate and robust predictive system that enables HR managers to identify burnout risk earlier and implement targeted interventions, supporting the organization’s transition toward proactive and data-driven workforce management.

7. Evaluation Metrics and Results

To properly evaluate the burnout prediction model, several performance metrics were used: Accuracy, Precision, Recall, and F1-Score. Each metric measures a different aspect of model performance and helps determine whether the model is suitable for real-world HR decision-making.

The second SVM training iteration utilized GridSearchCV-optimized hyperparameters (C, γ, kernel type) across five-fold cross-validation on the 38,508-sample SMOTE-balanced dataset, enhancing baseline 98.47% accuracy through refined decision boundary optimization and reduced false negatives critical for HR intervention reliability.

Accuracy measures the overall percentage of correct predictions made by the model. It is calculated by dividing the total number of correct predictions by the total number of predictions made. In this project, the optimized Support Vector Machine (SVM) achieved an accuracy of 98.47%, indicating that the model correctly classified most employee records. However, accuracy alone is not sufficient because the dataset originally had an 80/20 imbalance between burnout and non-burnout cases. In imbalanced datasets, a model may appear highly accurate even if it fails to detect the minority class.

Precision measures how many of the employees predicted to be at risk of burnout were actually at risk. High precision means the model produces fewer false positives, which is important because incorrect burnout predictions may cause unnecessary HR interventions or misallocation of resources. This metric ensures that when the system flags an employee as "At Risk," the prediction is reliable.

Recall measures how many of the actual burnout cases were successfully detected by the model. This metric is particularly critical for the organization because missing employees who are truly experiencing burnout (**false negatives**) can lead to reduced productivity, higher absenteeism, and employee turnover. A high recall ensures that most employees who need support are identified early.

F1-Score combines precision and recall into a single metric by calculating their harmonic mean. This metric is useful when both false positives and false negatives must be carefully balanced. In this study, the model achieved an F1-score of 0.96 for the Burnout class, demonstrating that it performs well in identifying at-risk employees while maintaining reliable prediction quality.

Overall, these evaluation metrics provide a comprehensive understanding of model performance. The high accuracy and strong F1-score indicate that the predictive model is capable of reliably identifying employees at risk of burnout, making it a valuable decision-support tool for HR teams. By using multiple metrics rather than relying on a single measure, the organization can confidently assess the model's quality and ensure it supports effective and informed workforce management decisions.

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